

Through-life Engineering Services (TES): (#2) Service Options (WHAT?)

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Sponsor: Bloc Digital/ATI/NATEP Published Date: 14 May 2024

Summary

Background

In this paper we expand upon the discussion of the Through Life Engineering Service (TES) value proposition¹, and address the questions of what form do these services typically take and how do they create the required service value?

The value proposition paper concluded that customers wanted more value creation (with less parasitic cost/value destruction) from engineered assets. This provides the opportunity to deliver through-life engineering services that either enable incremental value creation (more output) and/or reduce value destruction (less input or consequential cost).

Questions this paper addresses

- Where are Through life Engineering Services (TES) applied?
- What are the basic forms of TES?
- How is value released by TES?
- How can TES be contracted?
- What Through-life services should we consider (buying/supplying)?

Conclusion

Through-Life engineering services are typically applied where:

- the criticality of an asset (its reliable availability for use) and/or its costs of use (direct, indirect, consequential or risk) are high and there is a perceived opportunity to improve asset value creation or reduce asset influenced costs.
- A service provider (either internal or external to the user/owner organisation) is able to provide incremental gains through the application of their superior resources, knowledge or capabilities.

There are typically four basic service offering themes (usability, reliability, availability and value), beneath which a supplier may provide a number of individual service elements that enable these customer value outcomes.

¹ see *white paper (#1) : The Value Proposition*

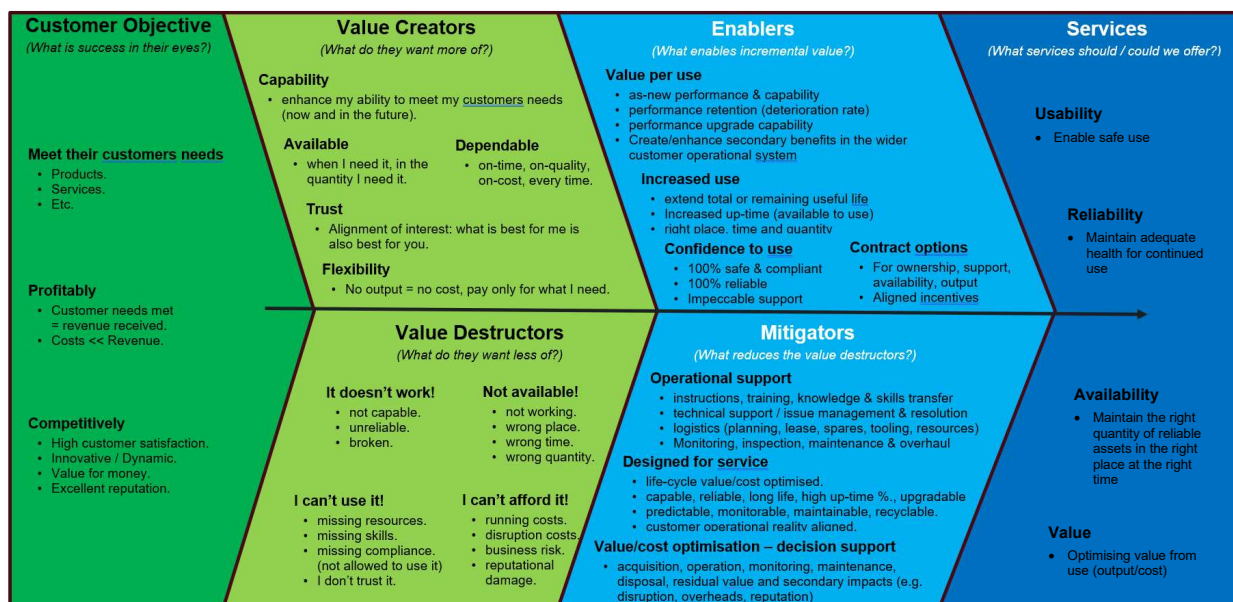


Figure 1: Value creation and enablement → service package themes

These service deliverables may in turn be contracted in various ways including but not limited to one or more of:

- pay per service event (invoice for each service activity provided),
- flat-rate/fixed price (e.g. annual or monthly service fee, providing an agreed level of service access),
- pay per use (e.g. fixed cost per hour of asset use),
- pay for availability (e.g. lease pool membership fee)
- pay per outcome (e.g. cargo tonne miles travelled)
 - possibly with service level guarantees and warranties for failed outcomes (e.g. 1000 service cost free miles per vehicle breakdown experienced).

The service themes, and the way in which they are contracted and warrantied, are often considered as a progression of ever greater transfer of risk and responsibility for the operation of assets to the service supplier, with usability as the basic service offer, progressing through to value as the most advanced service, and pay per event through to outcome-based contracting as the other risk transfer dimension.

However, it is common to find a degree of independence between the type of service being provided and the method of contracting. Often the contracting method may be tailored to the needs of individual customers, whilst the service delivered may be more uniform across customers. Customer preference for different contract types is likely to be impacted by their accountancy policies, cash flow considerations, balance sheet sensitivity and appetite for risk. Allowing a range of contracting methods may open the door to a wider range of potential customers.

The following table shows some of the service content and contracting styles commonly available in the provision of aircraft engine support contracts.

NOTE: when a service is packaged and contains a menu of individual service elements, there may be elements/combinations of several different contracting methods applied. Different contracting methods may also be applied to different parts of an assets architecture, or at different stages in the life-cycle of asset.

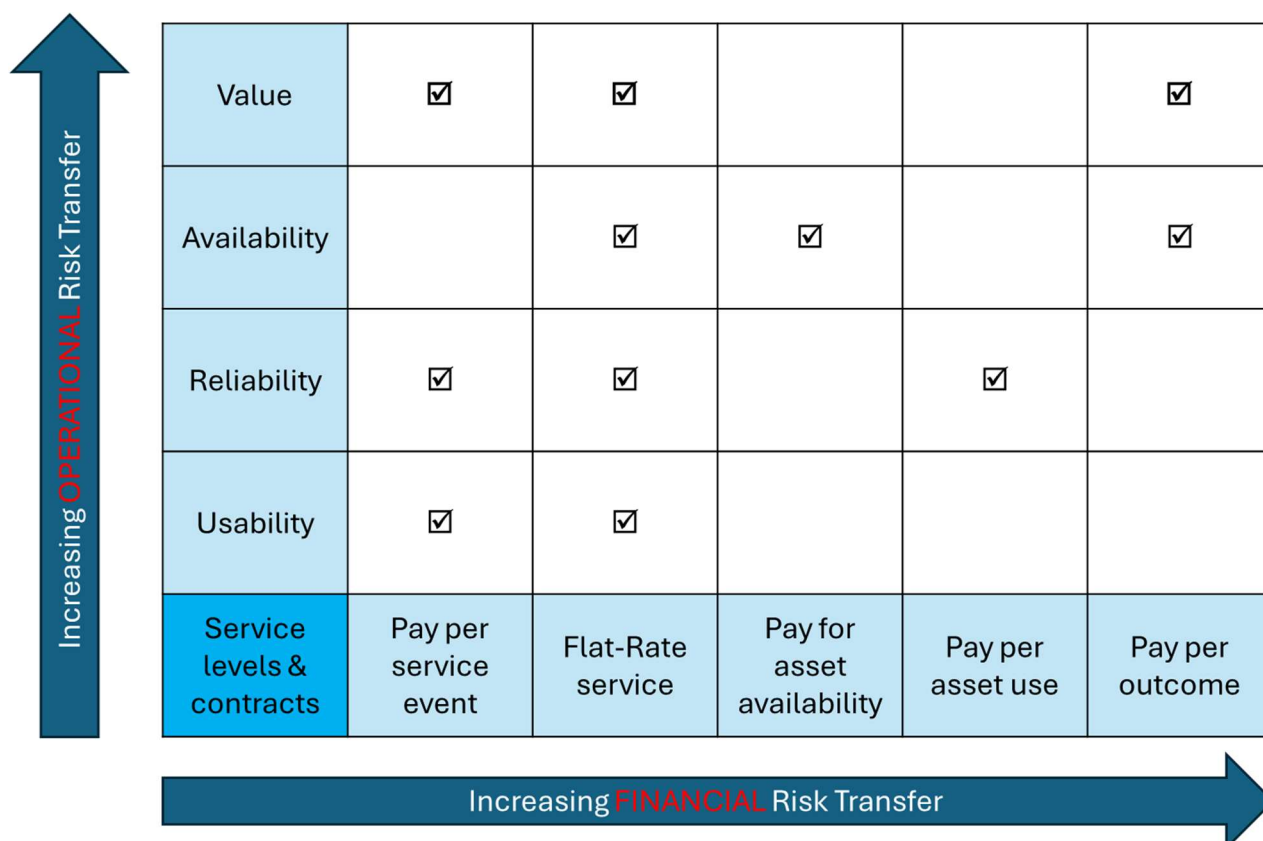


Figure 2: possible service content & contracting style combinations.

In order for any service theme (and its constituent parts) to add incremental value and to be commercially viable they have to pass a series of tests as an overall service package offer.

- **Risk** – is the perceived level of change (and hence risk) acceptable to both the supplier and customer?
- **Opportunity** – does the service package offer significant incremental opportunity for both the customer and supplier, and does the opportunity outweigh the risk?
- **Capability** – is the supplier credible, capable and competitive as a provider of the service?
- **Costs** – is the cost of change and ongoing service provision significantly less than the net opportunity (after risk provisions).

It is therefore important that in defining a service package to offer as a supplier, or to buy as a customer, the scope of the service and the contracting method is optimised for its commercial viability for both parties. This implies the need for increasing levels of mutual supplier/buyer understanding and trust, especially with the levels of operational and financial risk transfer associated with the more advanced service themes and contracting arrangements.

For this reason, many service providers offer a graduated menu of services, ranging from basic through to more advanced levels with varying forms of contracting terms. This allows new or existing customers to have options that match their current level of service value understanding and appetite for change / risk.

Equally, the nature of the asset, how it is used and where it is in its life-cycle can also drive the need for different service elements and contracting terms to be applied to different sections of the asset architecture, or to vary over its overall lifespan.

Content

Discussion

Where are TES applied?

As discussed in white paper (#1) - The Value Proposition:

Through-life engineering services are applied, where:

1. there is a perceived opportunity to improve value creation and/or reduce value destruction,
2. a service provider (as a result of better access to data, knowledge, capability or resources) is better placed to deliver the opportunity than the engineered assets owner/user.

To be commercially viable the service must:

- Address at least one significant area of opportunity.
 - high asset operational criticality (safety or dependence on output)
 - high cost of acquisition, ownership, use or disposal
 - high risk (reliability, availability, cost uncertainty, obsolescence etc)
- With significant levels of inefficiency (in the absence of a service)
 - e.g. lack of output, asset unreliability, lack of availability, excessive costs, operational disruption, loss of residual value etc
- Have a cost less than the perceived incremental value released.

What are the basic forms of TES?

TES can be offered at various levels, from the minimum possible offering to enable asset sales & use, through to offering guaranteed outcomes from the use of an asset, without the need for a customer to either own, use or manage it.

In the simplest form we can consider 4 basic levels of service:

- **Usability** – support to ensure that the asset can be used safely. Generally, this is a minimum requirement for supply of an asset and therefore contractually part of the initial asset sale. However, there may be opportunities for incremental safety or usability related services (on-site rather than remote support, enhanced or priority support, reliability not just safety support, proactive not just reactive, optional user or maintainer training etc)
- **Reliability** – the asset is inherently capable, robust and maintained, enabling it to be usable, for the required life-span, including preventative or reactive maintenance to retain or recover asset health.
- **Availability** – the right assets are in the right place, at the right time and the right quantity, with the right supporting resources to meet the operational need.
- **Value** – the asset, its ownership, and its use are optimised for outcomes (value/cost)

In practice these service levels form a progression staircase, with each service level forming a foundation for the next. Most organisations initially provide basic **usability** services and then progress through **reliability**, **availability** and potentially as far as **value** as their own competence, business opportunities and customer acceptance of the service value develop.

There also tends to be a progression in contracting method, with increasing levels of financial risk transfer to the service supplier, at the cost of a risk transfer premium. At the lowest level of risk transfer the consumer pays for the services they consume on a per event basis (e.g. paying for an annual car service event). At the other extreme the contract may be for the user outcomes achieved (e.g. per quality inspected item exiting a production line).

How is value released by TES?

The British Standards Institute published a publicly available specification in 2018 which outlines the principal value delivery streams of TES [*PAS 280:2018 Through-life engineering services. Adding business value through a common framework. Guide*].

The underlying thinking behind the framework is that an asset provides some level of required value adding capability to a user, that diminishes over time, through a combination of asset wear and tear and the availability of more modern and capable alternatives. TES therefore exists to maximise the duration and extent to which the asset can satisfy a users needs, and do so competitively.

The 4 value streams are:

1. AVOID – the causes of functionality, reliability or availability gaps (minimise functional deterioration, increase life, decrease maintenance down-time/cost or enable upgrade to lengthen the ability of the asset to compete with newer alternatives).
2. CONTAIN – when a functionality, reliability or availability gap is about to exist, or has occurred, minimise the user impact and any secondary consequences.
3. RECOVER – return the asset (or fleet of assets) to a healthy and capable state, that meets the required users needs for the next required period of time or use.
4. CONVERT – all experience gained in AVOID, CONTAIN & RECOVER activity into the ability to make incremental improvements in future decision making and execution, driving greater asset value delivery and/or lower asset created or influenced costs.

We can combine this with the previous basic forms of TES to begin to describe the elements of TES delivery and how they may be applied to the various levels of TES contract.

Note: the table below can be used to assess the potential for value adding Through-Life Services from either a customer or supplier perspective:

- CUSTOMER: are we receiving (or providing) these capabilities at an adequate quality, quantity and cost?
- SUPPLIER: are we better placed to provide these capabilities than the current (if any) supplier?

		Value stream			
		Avoid	Contain	Recover	Convert
Service Level	Value (value/cost optimisation)	Life cycle cost optimised design Asset financing/leasing Value/cost optimised guidance: - Usage - Maintenance timing, scope - Disposal & residual value recovery	Cost surveillance & opportunity alerting	Capability & life Upgrades Inspect/Repair innovations	Value/cost analytics, modelling & optimised decision support
	Availability (right assets, right place, right time, right quantity)	Availability optimised design (lifespan, up-time & maintainability) Availability risk optimised scheduling (inspection, maintenance, inventory, resources) Asset Leasing services Logistics planning	Spare assets, Lease pools & supply services Issue & campaign management	Transportation & storage Support infrastructure, tooling, resources, reactive maintenance planning	Availability risk analytics, modelling & optimised decision support
	Capability (healthy & functional assets)	Capable, robust design Smart assets (sensing, analysing & communicating) Risk optimised maintenance scheduling Inspection instructions Inspection schedule Inspection service Asset Health monitoring	Asset health alerting Fault diagnostics & troubleshooting Technical support	Maintenance instructions Spare parts supply Repair & Overhaul capabilities Remove & install service	Usage, health, failure risk and residual useful life analytics, modelling & optimised decision support
	Usability (Safe, usable assets)	Technical instructions for use Safety upgrades	Technical support Safety risk alerting	Safety issue management (investigate & resolve)	Safety analytics, modelling & optimised decision support

Table 1: Service Elements

In addition to the value streams PAS280 also describes the importance of planning.

In simple terms we can divide the various service value streams into 2 parts:

1. Defining what to do, when to do it and how much to do (Planning)
2. Executing the planned activities (Delivery)

In many cases the principal value proposition in a service is the ability to make the planning decisions, as once they are made >80% of the likely cost and benefit outcomes are probably fixed. A decision not to maintain an asset (whilst it still retains usable life) potentially generates significant additional output at zero cost, or alternatively generates 100% cost saving on the maintenance that would otherwise have been conducted.

There are case studies in the electronics world, where planning the repair of circuit boards rather than replacement can lead to assets returning to service years earlier than they would otherwise do (due to PCB supply lead times), at a 90% cost saving vs PCB replacement and at 20% of the environmental impact.

TES therefore releases its value by the choice of service being provided and the ability to generate a plan (data acquisition → analysis → planning decisions) and to then execute the plan. The capabilities required to deliver the plan/act sequence are discussed further in the 3rd paper in this series.

How can TES be contracted?

An implicit value of TES is the placement of risk with the party most capable of managing it. The customer pays a risk transfer premium) and the TES provider manages the risk and accepts its financial consequences.

The level of risk transfer will depend upon type of service being delivered, the level of guarantees and warranties associated with its delivery and the contractual framework used.

Dependent upon the maturity and capability of the TES provider, they may choose to offer varying levels of risk transfer, from the most limited (only providing enabling services, with minimal acceptance for their application and outcome risks), through to taking full ownership of the risk (and its financial consequences) including all the decisions and actions required to manage it.

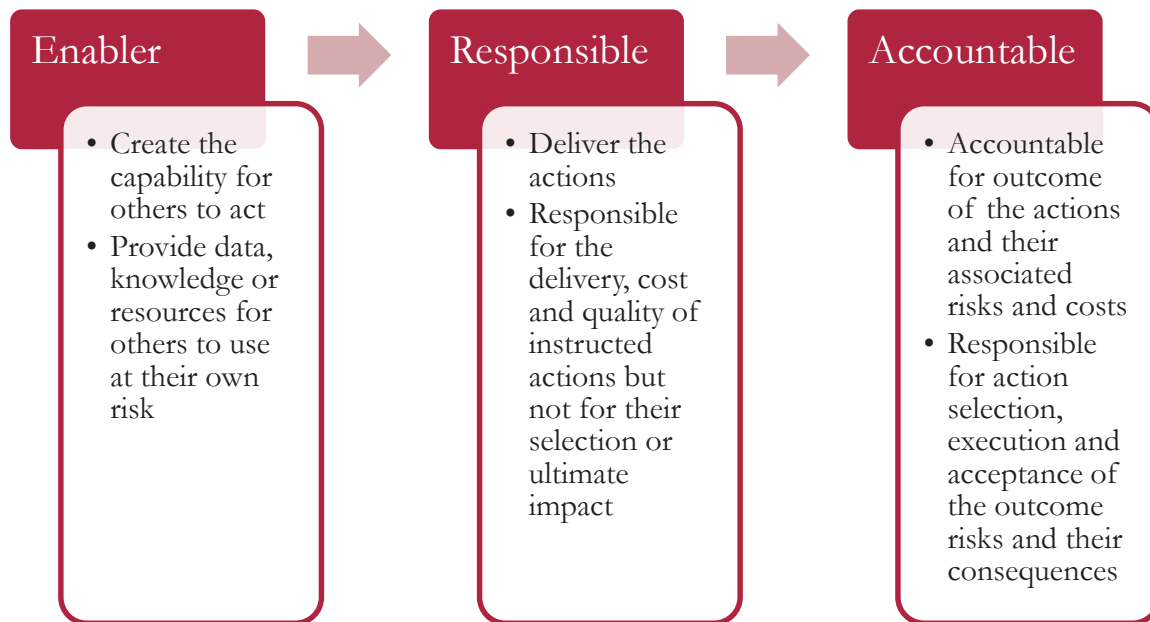
Dependent upon the maturity of the customer/supplier relationship, the customer may not wish to transfer all the risk, for fear that it will not be as well managed as the service offer advertises.

For this reason, recognising that different customers may have different risk appetites, many service providers offer a range of contractual methods that enable separation of some of the financial risk from the actual physical service delivered.

Contractual arrangements include:

- pay per service event (invoice for each service activity provided)
- flat-rate/fixed price (e.g. annual or monthly service fee, providing an agreed level of service access)
- pay per use (e.g. fixed cost per hour of asset use)
- pay for availability (e.g. lease pool membership fee)
- pay per outcome (e.g. cargo tonne-miles travelled)
 - possibly with service level guarantees and warranties for failed outcomes (e.g. 1000 service cost free miles per vehicle breakdown experienced).

Contracts will typically sit somewhere in a spectrum from **Enabler** through to **Accountable** as shown below.



Commercially the contracts may be executed on the basis of:

- Supply – minimal risk transfer
 - *Pay per service event supplied (physical parts, data, resource hours etc)*
 - *Flat-Rate service (fixed price for a service level agreement)*
- Availability – significant risk transfer, often in the form of guarantees and penalties
 - *Pay per item made available*
 - *Pay per period of availability*
- Usage – high risk transfer, in this contract form the highest TES supplier costs arise when the asset is not working, and therefore not generating any TES revenue. For the user, costs only occur when the asset is generating value for them.
 - *Pay per unit of use of an item*
- Outcome – maximum risk transfer
 - *Pay per unit of output from use*

Each TES contract will define a level of risk transfer to the supplier, and an aligned commercial agreement. It is critical to success that the contracting arrangement places both the financial accountability for risk and the ability/freedom to manage it with the same party.

Factors to consider when selecting a commercial framework for a service include:

- **RISK:** Does the contract place both control and ownership of the risk consequences with the same party (so that they can manage their risk exposure)?
 - Including data access and decision rights
- **REWARD:** Does the contract reward all stakeholders equitably and sustainably?
- **ALIGNMENT:** Does the contract align motivational drivers for the service provider and customer, increasing the likely delivery of overall value/cost outputs?
- **CASH FLOW/BALANCE SHEET:** Does the contracting method generate changes in cash flow profile and/or balance sheet assets for either customer or supplier, and if so, is the change beneficial, detrimental or negligible (e.g. for pay per use contracts: are customer service cost cash flows and volatility better aligned with their revenue generation)?
- **BURDEN:** Does the contracting method generate more or less administrative burden (e.g. usage data capture/validation, number of invoice/payment transactions required, auditing etc)

What Through-life services should we consider (buying/supplying)?

The approach to prioritising TES elements can be simplified to a number of key steps, using table 1 as a framework for evaluation:

1. RISK: Where are we starting from?

In most cases the decision on what service level to consider, and how to contract it is predicated upon the current position and the level of change it requires. Change involves both cost and risk, which for many situations will dictate an incremental approach to the evaluation, piloting and deployment of services. The perceived level of cost and risk change will have to be acceptable to both the supplier and customer, requiring a good understanding of their current position and tolerance/appetite for change.

2. OPPORTUNITY: Is there a significant improvement (in net value) opportunity?

For each of the service elements, are they currently operating at a level which offers minimal, significant or high incremental value opportunity if they were improved/implemented?

3. CAPABILITY: Is the potential TES provider capable of delivering the value opportunity?

Is the proposed provider of the service element the 'best' placed to provide it (knowledge, capability, capacity, quality, cost etc), or are alternative suppliers more likely to deliver incremental value more effectively?

4. COST: Is the cost/risk of change acceptable?

Is the cost of implementation and operation of the service element minimal, significant or high?
Is the commercial contracting method aligned to the service being delivered? Will it incentivise the right behaviours in the right stakeholders and place the freedom to control risk with the party who has holds the risk consequences?

Potential service elements to enable/adopt can then be prioritised based upon:

- their individual risk/opportunity/capability and cost assessments
- their interdependence (recognising that some service elements enable/require others)

Consideration should be given to packaging TES change into incremental chunks, where levels of risk and cost can be evaluated, measured and controlled to acceptable levels through pilot implementations prior to full scale deployments.

Input Received

This paper has been funded by the InspectorPlus research programme (Bloc Digital, ATI, NATEP, University of Derby, DFS-Consulting).



The content is based upon the combined experience of many industrial and academic organisations, shared through membership and activities sponsored by the TES Council and other collaborative research and industrial peer group activities (including authoring the British Standards Institute guidance **PAS-280 – Through-life Engineering Services** and the creation of the UK national apprentice standard **Through-life engineering specialist - integrated degree - level7**)

